Shanza Zahir

AI/ML Engineer|API & Backend Development
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SUMMARY

AI/ML engineer born and raised in Abu Dhabi, UAE with a First Class BSc Hons in Software Engineering from Liverpool John Moores University, UK and six industry roles. Built API integrations on Apigee for banking/government clients all over Gulf countries at iSolutions and built RAG chatbots and CPU-optimized AI models at Bitknox. Two hackathon builds in multi-agent systems and vector search. Open to remote roles.

EXPERIENCE

API Developer | **iSolution (Google Cloud Partner) Riyadh, Saudi Arabia** **May 2026 – Nov 2026**

- Learned and applied Apigee on GCP: built API proxies and implemented OAuth, security policies, transformations and mappings.
- Debugged proxy errors and ran rigorous proxy testing before release; worked on WSO2 mappings and IBM webMethods integrations. Delivered integrations for banking and government clients across the Gulf region.
- Joined client calls to diagnose issues, agreed resolutions with them and wrote the technical documentation.

AI/ML Engineer | **Bitknox Global** **Jan 2026 – May 2026**

- Built text-to-image pipelines with Stable Diffusion, optimized for Intel CPUs via OpenVINO/ONNX, with latent caching for low-latency inference.
- Integrated PGVector with PostgreSQL for RAG chatbots; developed an AI image search library for web integration.
- Developed URDF models and simulations in Gazebo Harmonic, PyBullet and ROS2 Jazzy; implemented the Unitree SDK and researched HarmonyOS for humanoid robotics R&D.
- Architected a lead-gen portal for automated Google Maps scraping and WhatsApp/email campaigns (FastAPI, Tailwind, Playwright, MySQL).
- Authored the architecture for a 150k-user field-staff tracking project for a provincial government; delivered viability reports and monitoring tools.

Game Design & QA Intern | **The Game Storm Studios** **Oct 2025 – Dec 2025**

- Ran stress-testing and QA cycles on 42 games, uncovering 270+ bugs; designed 7 full-scale Game Design Documents.

Freelance IT Consultant & Tutor | Remote **Jan 2025 – Mar 2025**

- Taught cloud computing, virtualization and embedded systems; helped design high-availability architectures with Docker, MySQL NDB clusters, MongoDB replicas and Nginx load balancing.

IT Intern | **DHA Main Office** **Oct 2024 – Dec 2024**

- Helped roll out Jira for project management: research, feature testing, creating projects, reports and presentations.

ML Engineer Intern, AWS Cloud Services | **NetSol Technologies** **Jul 2024 – Sep 2024**

- Worked on generative AI projects with large language models; collaborated with senior developers on model training, validation and fine-tuning best practices.

HACKATHON PROJECTS

Dispatch | **Google AI Hackathon**

Multi-agent AI command system for emergency dispatchers: understands calls in any language/dialect, produces a structured incident and response plan for the dispatcher to approve or reject and alerts only nearby citizens. Built with Expo, Node.js, multi-agent orchestration and Google Antigravity.

Chemical Symbiosis | Google AI Seekho 2026 Hackathon

AI matchmaker that turns one factory's waste into another's raw material: gemini-embedding-001 with Qdrant semantic search, Gemini 2.5 Flash to explain the chemistry; deployed on Google Cloud Run.

EDUCATION

BSc (Hons) Software Engineering, First Class (75%) | **Liverpool John Moores University, Liverpool, England** **Sept 2021 – Sept 2024**

- 2× High Achiever Award

SKILLS

Cloud & APIs: Apigee, Google Cloud (Cloud Run, ADK), WSO2, IBM webMethods, Docker
AI/ML: RAG, LLMs, multi-agent systems, OpenVINO, ONNX, TensorFlow, scikit-learn
Backend & data: Python, Java, JavaScript, Node.js, SQL, FastAPI, PostgreSQL/PGVector, Qdrant, MySQL, MongoDB, Power BI, Tableau
Robotics & embedded: ROS2, Gazebo, PyBullet, URDF, C/C++, ESP32
Languages: English, Urdu, Arabic (basic)

CERTIFICATES

IBM Machine Learning Professional Certificate **|** Google Business Intelligence Certificate **|** DeepLearning.ai Machine Learning and Deep Learning Specialization **|** Google Data Analytics

Google Cloud Skills Boost (2026): Engineer AI Agents with Agent Development Kit **|** Build Serverless Applications with Cloud Run Functions **|** Introduction to Large Language Models **|** API Security and API Design with the Apigee API Platform.